PERFECT OBJECTS ARE COMPACT ON A SITE WITH QUASI-COMPACT OBJECTS AND FINITE COVERINGS, AND ON THE BOUNDED DERIVED CATEGORY ALWAYS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reformulation through the dual, and the two applications of the compactness criteria with the commutation of cohomology with direct sums.

Editorial note. The label lemma-perfect-is-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the four cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The derived dual and its tensor description of Hom. Same chapter, the lemma identifying Hom(K, M) with H^0 of the tensor product of M with the dual of the perfect object K. Supplies the reformulation of compactness as commutation of the derived global sections with direct sums.
- Compactness of the extension by zero of the structure sheaf. Same chapter, the two lemmas giving compactness of j_{U!}O_U under bounded cohomological dimension and in the bounded below setting. Supplies the criteria applied after the reformulation, with U = X.
- Commutation of cohomology with direct sums. Cohomology on Sites, the lemma computing the cohomology of a direct sum over a collection, applicable by the assumptions on quasi-compact objects and coverings. Supplies the commutation of H^p(X, -) with direct sums in part (b).

Source reference. Original reference 0739: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 2518–2518. Prerequisite pointer invoked at proof line(s) 14225; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07VI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13048–13048. Prerequisite pointer invoked at proof line(s) 14208; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 094D: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14136–14136. Prerequisite pointer invoked at proof line(s) 14227; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0G22: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14109–14109. Prerequisite pointer invoked at proof line(s) 14220; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Assume $\mathcal{C}$ has the following properties*

(1) *$\mathcal{C}$ has a quasi-compact final object X,*
(2) *every quasi-compact object of $\mathcal{C}$ has a cofinal system of coverings which are finite and consist of quasi-compact objects,*
(3) *for a finite covering $\{U_i \to U\}_{i \in I}$ with U, U_i quasi-compact the fibre products $U_i \times_U U_j$ are quasi-compact.*

Let K be a perfect object of $D(\mathcal{O})$. Then

(a) *K is a compact object of $D^+(\mathcal{O})$ in the following sense: if $M = \bigoplus_{i \in I} M_i$ is bounded below, then $\mathrm{Hom}(K, M) = \bigoplus_{i \in I} \mathrm{Hom}(K, M_i)$.*
(b) *If $(\mathcal{C}, \mathcal{O})$ has finite cohomological dimension, i.e., if there exists a d such that $H^i(X, \mathcal{F}) = 0$ for $i > d$ for any $\mathcal{O}$-module $\mathcal{F}$, then K is a compact object of $D(\mathcal{O})$.*

Proof. Let $K^\vee$ be the dual of K, see Lemma 07VI. Then we have

$$\mathrm{Hom}_{D(\mathcal{O})}(K, M) = H^0(X, K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}} M)$$

functorially in M in $D(\mathcal{O})$. Since $K^\vee \otimes^{\mathbf{L}}_{\mathcal{O}} -$ commutes with direct sums it suffices to show that $R\Gamma(X, -)$ commutes with the relevant direct sums.

Proof of (a). After reformulation as above this is a special case of Lemma 0G22 with $U = X$.

Proof of (b). Since $R\Gamma(X, K) = R\,\mathrm{Hom}(\mathcal{O}, K)$ and since $H^p(X, -)$ commutes with direct sums by Lemma 0739 this is a special case of Lemma 094D. □